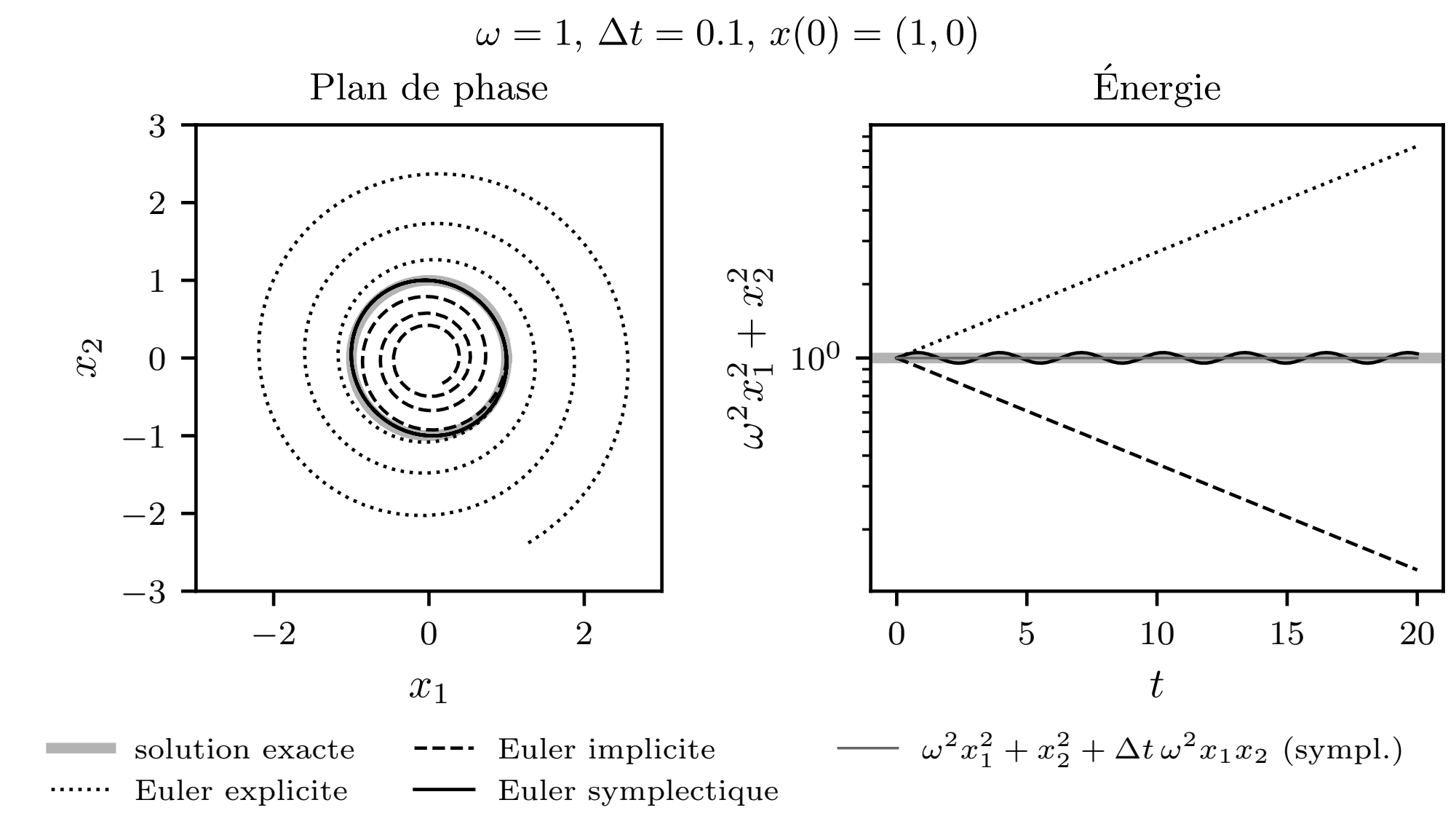

$\omega = 1,\ \Delta t = 0.1,\ x(0) = (1, 0)$
Plan de phase
Énergie
x_2
x_1
$\omega^2 x_1^2 + x_2^2$
t
10^0
3
2
1
0
−1
−2
−3
−2
0
2
0
5
10
15
20
solution exacte
Euler explicite
Euler implicite
Euler symplectique
$\omega^2 x_1^2 + x_2^2 + \Delta t\, \omega^2 x_1 x_2$ (sympl.)